Syllabus Year: 2023-2024

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - II

Course Code: BCA104-2 **Max Marks: 100**

Course Name: DISCRETE MATHEMATICS **Duration: 03:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Demonstrate a working knowledge of set notation and elementary set theory and recognize the connection between set operations and logic.

CO2: Apply the different properties of injections, surjections, bijections, compositions, and inverse functions.

CO3: Understand different types of matrices and operations among matrices.

CO4: Determine when a relation is reflexive, symmetric, antisymmetric, transitive, apply the properties of equivalence relations and partial orderings, and explain the connection between equivalence relations.

CO5: Effective express the elementary concepts and results of Number theory.

Section A **3*8=24**

1)	Determine the smallest positive integer that leaves a remainder of 4 when divided by 7 and a remainder of 3 when divided by 9.	**3**	(CO1, L3)
2) a)	Explain with an example how you would determine if a given function is bijective and describe its inverse function.	**3**	(CO2, L2)
	OR		
b)	Explain how to determine if a function is an injection and provide an example to illustrate your explanation.	**3**	(CO2, L2)
3)	Analyze and differentiate between an upper triangular matrix and a diagonal matrix using examples.	**3**	(CO3, L4)
4)	Determine if the relation R = {(1, 1), (2, 2), (3, 3), (1, 2), (2, 1), (2, 3)} on the set A = {1, 2, 3} is symmetric and provide a one-line explanation for your answer.	**3**	(CO4, L3)
5) a)	Explain what it means for a function to be bijective and provide an example of a bijective function from the set {1, 2, 3} to itself.	**3**	(CO5, L2)
	OR		
b)	Discuss how the concept of congruence modulo n helps in determining if an integer belongs to the same equivalence class as another integer under a given modulus. Provide an example.	**3**	(CO5, L2)
6)	Analyze and differentiate between the Euclidean algorithm and the method of successive subtraction for finding the greatest common divisor (GCD) of two integers, providing a small example for each to illustrate your explanation.	**3**	(CO1, L4)

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7) a)	Determine if the function f: Z → Z defined by f(x) = x + 5 is an injection, surjection, or bijection. Justify your answer.	**3**	(CO2, L3)
	OR		
b)	Determine if the function : → defined by () = 2 + 1 is an injection, surjection, or bijection when = {0, 1, 2} and = {1, 3, 5}. Justify your answer.	**3**	(CO2, L3)
8)	Determine whether the set of all 2x2 matrices with integer entries is countable or not. Justify your answer.	**3**	(CO3, L3)
	Section B		**8*7=56**
9)	Consider the set = {1, 2, 3, 4} and relations 1 × defined by 1 = {(1, 1), (2, 2), (3, 3), (4, 4), (1, 2), (2, 1), (2, 4), (4, 2)}. Determine whether 1 is reflexive, symmetric, antisymmetric, and transitive. If not, provide a counterexample for each property that fails. Additionally, construct an equivalence relation from 1 if possible, by adding the minimum number of elements to make it an equivalence relation. Show all your work.	**8**	(CO4, L3)
10) a)	Analyze the properties of the given relation R defined on the set A = {1, 2, 3, 4} as R = {(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (3, 3), (4, 4)}. Determine whether this relation is symmetric and justify your answer. Further, consider the function f: A → B defined by f(x) = x^2 - 5x + 6 for all x A where B is the set of integers. Test if f is bijective by examining its injectivity and surjectivity properties.	**8**	(CO5, L4)
	OR		
b)	Analyze the divisibility relation defined over the set = {1, 2, 3, 4, 5} and prove whether it forms a partial order by verifying reflexivity, antisymmetry, and transitivity. Additionally, categorize each pair (,) in as symmetric or asymmetric. Finally, determine if the set of integers under this divisibility relation satisfies the properties of an equivalence relation.	**8**	(CO5, L4)
11)	Discuss the concept of countability in sets and prove that both the set of rational numbers and the Cartesian product of two countable sets are also countable.	**8**	(CO1, L2)
12) a)	Determine if the function : → defined by () = 2 + 1 is an injection, a surjection, or both. Justify your answer using properties of functions and provide examples to support your claims. Additionally, construct the inverse relation −1 if is invertible, and determine if is a function.	**8**	(CO2, L3)
	OR		
b)	Given two functions, f: A → B defined by f(x) = 2x + 3 and g: B → C defined by g(y) = y^2 - 1, where A = {0, 1, 2}, B = {3, 5, 7}, and C = {8, 4, 0}. Determine (i) the composition of functions f g, (ii) whether f is an injection, surjection, or bijection, (iii) if a function h: A → C can be defined as h(x) = x^2 + 1, determine if h is a bijection. Provide justifications for your answers.	**8**	(CO2, L3)
13)	Analyze the matrix representations of two different 4x4 matrices corresponding to relations 1 and 2 on a set = {a, b, c, d} defined as follows: Matrix for 1: [1 0 1 0] [0 1 1 0] [1 0 1 1] [0 0 1 1] Matrix for 2: [1 0 0 1] [1 1 1 0] [0 1 1 1] [0 0 1 1] (i) Determine if each relation is reflexive, symmetric, or transitive. Justify your answers. (ii) Find the matrix representing the composition 2 1 and describe what this composition represents in terms of the relations. (iii) Construct a new relation 3 that includes all elements from both 1 and 2, and determine if 3 is symmetric. Explain your reasoning.	**8**	(CO3, L4)

14) a)	Discuss the properties of an equivalence relation on a set of integers. Consider the relation R defined by xRy if and only if x ÷ y = k, where k is a non-zero integer. (i) Explain whether R is reflexive, symmetric, or transitive. (ii) If R is found to be an equivalence relation, describe its equivalence classes. (iii) Discuss the connection between this relation and modular arithmetic using specific examples.	**8**	(CO4, L2)
	OR		
b)	Discuss the properties of equivalence relations and partial orderings with respect to their definitions, and explain how these concepts are related in the context of modular arithmetic. Provide examples to illustrate your points.	**8**	(CO4, L2)
15)	Solve the following system of linear congruences using the Chinese Remainder Theorem: (i) Determine if the given congruences have a solution. (ii) If a solution exists, find it and express your answer in the form ≡ ( ), where is the product of moduli from the system. (iii) Verify that your solution satisfies all given congruences.	**8**	(CO5, L3)
	Section C		**10*2=20**
16)	Consider two sets: = {1, 2, 3, 4} and = {1, 2, 3}. Let there be relations and on set such that = {(1, 1), (2, 2), (3, 3), (4, 4), (1, 2), (2, 1)}, = {(1, 1), (2, 2), (3, 3), (4, 4), (2, 3), (3, 2)}. (i) Determine whether and are symmetric relations. Justify your answer using the definitions of symmetry. (ii) Calculate the intersection ∩ and the union . Represent each operation in terms of ordered pairs. (iii) Construct a relation matrix for both and , then find the matrix representing the symmetric difference ( ). Explain your steps.	**10**	(CO1, L3)
17) a)	Consider two functions : → and : → defined by () = 2 + 1 for , where = {0, 1, 2}, = {1, 3, 5}, and () = ^2 - 1 for . Examine whether the composition of these functions, denoted as ( ), is a bijection or not. If it is a bijection, find its inverse function ( )^(-1). Additionally, discuss if there exists an injection from set to set through the given composition and justify your answer.	**10**	(CO2, L4)
	OR		
b)	Examine and prove whether the union of two countable sets, A and B, is also countable. Additionally, differentiate between injections, surjections, and bijections in the context of these mappings between the sets.	**10**	(CO2, L4)

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember
L2 - Understand
L3 - Apply
L4 - Analyse
L5 - Evaluate
L6 - Create

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